

ADB LOGCAT – REAL-TIME LOGS MONITOR

◇ Introduction

Logcat Android ka built-in logging system hai jo system aur applications ki real-time logs display karta hai. ADB ke zariye aap command line se logs monitor, filter aur save kar sakte hain. Ye debugging aur troubleshooting ke liye sabse important tools mein se ek hai.

◇ REAL-TIME LOGS DEKHNA

Command

```
adb logcat
```

Description

Ye command Android device ki saari logs real-time mein display karti hai. Output continuously screen par aata rehta hai jab tak aap **Ctrl + C** press karke process stop nahi karte.

◇ LOGS KO FILTER KARNA

1. Priority Level ke Hisaab Se Filter

Har log ka ek priority level hota hai jo uski importance batata hai.

Verbose (V) – Sabse zyada detailed logs.

Debug (D) – Debugging information.

Info (I) – General information.

Warning (W) – Warning messages.

Error (E) – Error messages.

Fatal (F) – Critical ya fatal errors.

Silent (S) – Kisi bhi log ko display nahi karta.

Example – Sirf Error Logs

```
adb logcat *:E
```

Ye command sirf **Error** aur **Fatal** level ki logs display karegi.

Example – Warning aur Uske Upar

```
adb logcat *:W
```

Is command se **Warning, Error aur Fatal** level ki logs dikhengi.

◇ TAG KE HISAAB SE FILTER

Har log ka ek **Tag** hota hai jo batata hai ki log kis application ya system component se generate hui hai.

Syntax

```
adb logcat <Tag>:<Priority> *:S
```

Example – Sirf MyApp ki Debug Logs

```
adb logcat MyApp:D *:S
```

Ye command sirf **MyApp** tag ki Debug aur usse higher priority ki logs display karegi. *:S baaki sab logs ko hide kar deta hai.

Example – Multiple Tags

```
adb logcat ActivityManager:I MyApp:D *:S
```

Ye command **ActivityManager** ki Info logs aur **MyApp** ki Debug logs ek saath display karegi.

◇ GREP KE ZARIYE FILTER (ADVANCED)

Agar aap keyword ke basis par logs dekhna chahte hain to **grep** command use kar sakte hain.

Example – Keyword Search

```
adb logcat | grep "error"
```

Ye command sirf un logs ko display karegi jinmein **error** word maujood hoga.

Example – Case Insensitive Search

```
adb logcat | grep -i "myapp"
```

Ye command uppercase aur lowercase dono ko ignore karke search karegi.

Example – Tag ke Hisaab Se Search

```
adb logcat | grep "^..MyApp"
```

Ye command sirf **MyApp** tag wali logs display karegi.

◇ SPECIFIC APP KI LOGS (PACKAGE NAME SE)

Agar sirf ek application ki logs dekhni hon to uska **PID (Process ID)** use kiya ja sakta hai.

Method 1 – PS + Grep

```
adb logcat | grep "$(adb shell ps | grep  
com.example.app | awk '{print $2}')
```

Ye command pehle app ka PID find karti hai aur phir us process ki logs display karti hai.

Method 2 – PID Flag

```
adb logcat --pid=12345
```

Agar PID pehle se pata ho to ye command sirf us process ki logs dikhayegi.

Method 3 – PIDCAT Tool

```
pidcat com.example.app
```

`pidcat` ek Python-based utility hai jo package name ke basis par sirf us application ki colored aur readable logs display karti hai.

◇ LOG OUTPUT FORMAT CHANGE KARNA

`-v` option ka use log output ka format change karne ke liye hota hai.

brief – Default format. Priority, Tag aur PID display karta hai.

time – Date aur time ke saath logs display karta hai.

threadtime – Date, time, thread ID (TID), process ID (PID) aur tag sab show karta hai.

raw – Sirf message display karta hai, metadata nahi.

long – Complete metadata ke saath detailed output show karta hai.

Example

```
adb logcat -v threadtime
```

Ye command time aur thread information ke saath logs display karegi.

◇ LOGS KO FILE MEIN SAVE KARNA

Method 1 – Output Redirection

```
adb logcat > logcat.txt
```

Saari logs `logcat.txt` file mein save ho jayengi. Recording stop karne ke liye **Ctrl + C** press karein.

Method 2 – Time Format ke Saath

```
adb logcat -v time > logs.txt
```

Ye command logs ko timestamp ke saath file mein save karegi.

Method 3 – Device Par Save Karna

```
adb logcat -f /sdcard/logs.txt
```

Ye command logs ko directly Android device ki storage mein save karegi.

Method 4 – Sirf Crash Logs Save Karna

```
adb logcat -b crash > crash_logs.txt
```

Ye command sirf crash buffer ki logs ko file mein save karegi.

◇ LOG BUFFER CLEAR KARNA

Command

```
adb logcat -c
```

Ye command Logcat buffer ki purani logs ko clear kar deti hai.

◇ DIFFERENT LOG BUFFERS

Android mein alag-alag log buffers available hote hain.

main – Default application logs.

system – Android system logs.

events – System event logs.

crash – Application crash logs.

radio – Radio aur telephony related logs.

Example – Sirf Crash Logs

```
adb logcat -b crash
```

Ye command sirf crash buffer ki logs display karegi.

◇ QUICK REFERENCE COMMANDS

Real-time logs dekhne ke liye:

```
adb logcat
```

Sirf Error logs dekhne ke liye:

```
adb logcat *:E
```

Sirf MyApp ki Debug logs dekhne ke liye:

```
adb logcat MyApp:D *:S
```

Keyword search ke liye:

```
adb logcat | grep "error"
```

Thread aur time ke saath logs dekhne ke liye:

```
adb logcat -v threadtime
```

Logs ko file mein save karne ke liye:

```
adb logcat > logs.txt
```

Log buffer clear karne ke liye:

```
adb logcat -c
```

Sirf crash logs dekhne ke liye:

```
adb logcat -b crash
```

Specific application ki colored logs dekhne ke liye:

```
pidcat com.example.app
```

IMPORTANT NOTE

Ye notes sirf educational aur learning purposes ke liye hain. Kisi bhi device par bina permission ke access, monitoring ya debugging karna IT Act 2000 aur anya applicable cyber laws ke under punishable offense ho sakta hai.

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